The Prime Side Does Not Compress

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Abstract

The archimedean side of the Weil explicit formula, expanded in the Meixner–Pollaczek basis at $\lambda = \frac{1}{4}$, lies in a rank-two lattice: every entry is $\mathbb{Q}\sqrt{2\pi} + \mathbb{Q}\sqrt{2\pi}\,(\gamma + \log 8\pi)$. We compute the prime side in the same basis in closed form — each entry is an explicit $\mathbb{Q}$-linear series in $-\zeta'/\zeta(3), -\zeta'/\zeta(5), \ldots$ — and show numerically, to 500 digits, that no analogous compression occurs: the sixteen entries with $m, n \le 6$ are $\mathbb{Q}$-linearly independent of each other and of the archimedean lattice. The two halves of the explicit formula are algebraically incommensurable in this basis. The integrality of the approximation matrix is purely archimedean.

1 The quadratic form

Let $\widehat{P}_m$ denote the normalized Meixner–Pollaczek polynomials at $\lambda = \frac{1}{4}$, $\phi = \frac{\pi}{2}$, with $h_m = 2\pi\,\Gamma(m + \frac{1}{2})/(\sqrt{2}\,m!)$. Polarizing the Weil quadratic form on the basis $F_m = \Gamma(\frac{1}{4} + ix)\widehat{P}_m$ and regularizing the pole at $s = 0, 1$ gives the kernels

$$h_{mn}(r) \;=\; \big(r^2 + \tfrac{1}{4}\big)\,\widehat{P}_m\big(\tfrac{r}{2}\big)\widehat{P}_n\big(\tfrac{r}{2}\big)\,\big|\Gamma\big(\tfrac{1}{4} + \tfrac{ir}{2}\big)\big|^2, \qquad m \equiv n \pmod 2.$$

The MP weight is the Weil kernel. The explicit formula (validated to 21 digits against the first 60 zeros) reads

$$W_{mn} \;=\; \sum_{\gamma} h_{mn}(\gamma) \;=\; \underbrace{2\,\Gamma(\tfrac{1}{2})\Big[\widehat{P}_m\widehat{P}_n\big(\tfrac{i}{4}\big) + \widehat{P}_m\widehat{P}_n\big(-\tfrac{i}{4}\big)\Big]}_{\text{pole}} \;+\; \underbrace{\frac{1}{2\pi}\int_{\mathbb{R}} h_{mn}(r)\Big[\operatorname{Re}\psi\big(\tfrac{1}{4} + \tfrac{ir}{2}\big) - \log\pi\Big]dr}_{\text{arch}} \;-\; P_{mn},$$

where $P_{mn} = 2\sum_n \Lambda(n)\,n^{-1/2} g_{mn}(\log n)$ and g_{mn} is the Fourier transform of h_{mn}. The factor $r^2 + \frac{1}{4}$ does not kill the pole terms: zero times Γ-pole leaves the finite residue shown.

The three known sides compress as follows. The pole side is $\mathbb{Q}\sqrt{\pi}$ exactly (the $\widehat{P}_m(\pm\frac{i}{4})$ are rational up to the norms). For the archimedean side, the identity $2\operatorname{Re}\psi \cdot |\Gamma|^2 = \partial_\lambda |\Gamma(\lambda + ix)|^2$ reduces every entry to MP moments, giving:

Theorem 1.1. *For all $m \equiv n \pmod 2$,*

$$\sqrt{h_m h_n}\cdot \operatorname{arch}_{mn} \;\in\; \mathbb{Q}\,\sqrt{2\pi} \;\oplus\; \mathbb{Q}\,\sqrt{2\pi}\,(\gamma + \log 8\pi).$$

For example $\sqrt{h_0}\,h_0^{1/2}\operatorname{arch}_{00} = \sqrt{2\pi}\,[\,1 - \frac{3}{4}(\gamma + \log 8\pi)\,]$, and the entries $(0,4)$, $(0,6)$, $(1,5)$, $(2,6)$ are pure rational multiples of $\sqrt{2\pi}$: $\frac{3}{16}$, $-\frac{1}{16}$, $\frac{15}{32}$, $\frac{97}{128}$.

2 The prime side in closed form

Theorem 2.1. *Let $t = \tanh u$. Then*

$$g_{mn}(u) \;=\; \frac{\sqrt{2\pi}}{\sqrt{h_m h_n}}\,T_{mn}(\tanh u)\,\operatorname{sech}^{1/2}(u), \qquad T_{mn} \in \mathbb{Q}[t], \quad T_{mn}(1) = 0.$$

Proof. The Fourier transform of the weight is $\frac{1}{2\pi}\int |\Gamma(\frac{1}{4}+\frac{ir}{2})|^2 e^{-iur}\,dr = \sqrt{2\pi}\,\mathrm{sech}^{1/2}(u)$. Multiplication by r corresponds to $i\,d/du$, and d/du acts on $p(t)\,\mathrm{sech}^{1/2} u$ by $p \mapsto (1-t^2)p' - \frac{t}{2}p$, so the polynomial factor $(r^2+\frac{1}{4})P_mP_n$ becomes a differential operator preserving $\mathbb{Q}[t]\,\mathrm{sech}^{1/2} u$. The vanishing $T_{mn}(1)=0$ is the zero of $r^2+\frac{1}{4}$ at $r=-\frac{i}{2}$, which kills the leading $e^{-u/2}$ decay. E.g. $T_{00} = \frac{3}{4}(1-t^2)$. □

At $u = \log n$ everything is algebraic in n: $\tanh(\log n) = \frac{n^2-1}{n^2+1}$, $\mathrm{sech}(\log n) = \frac{2n}{n^2+1}$. Writing $w = n^{-2}$ and expanding $\psi_{mn}(w) = (1+w)^{-1/2}\,T_{mn}\big(\frac{1-w}{1+w}\big) = \sum_{k\ge1} a_k w^k$ with $a_k \in \mathbb{Q}$ ($a_0 = 0$ by $T_{mn}(1)=0$), the von Mangoldt sum collapses onto ζ'/ζ at odd integers:

Corollary 2.2.

$$P_{mn} \;=\; \frac{4\sqrt{\pi}}{\sqrt{h_m h_n}} \sum_{k\ge1} a_k^{(mn)} \left(-\frac{\zeta'}{\zeta}(1+2k)\right), \qquad a_k^{(mn)} \in \mathbb{Q}.$$

The terms decay like 4^{-k}: five hundred digits cost 856 terms. For example

$$P_{00} = \frac{6\sqrt{2}}{\pi} \sum_{k\ge1} \binom{-5/2}{k-1} \left(-\frac{\zeta'}{\zeta}(1+2k)\right) = 0.31111185543474192\ldots$$

Without the vanishing $T_{mn}(1)=0$ the terms would be $\Lambda(n)/n$ and the sum would diverge; with it they are $\Lambda(n)/n^3$, and direct summation to 40 digits would still require a cutoff near 10^{20}. The series above is the only practical route.

3 The negative result

Observation 3.1. At 500 digits of working precision, with $A = \sqrt{2e}\,K_0(\frac{1}{2})$, $B = \sqrt{2e}\,K_1(\frac{1}{2})$:

1. no entry P_{mn}, $m,n \le 6$, satisfies a $\mathbb{Z}$-relation with $\{A,B\} \times \{1, \log 2, \log\pi, \pi, \gamma\}$ (PSLQ, coefficients to 10^{10}, sixteen entries);
2. the sixteen entries admit no $\mathbb{Q}$-relation among themselves: their $\mathbb{Q}$-rank is 16;
3. the single candidate relation found at coefficient bound 10^{14} had all coefficients $\sim 10^{13}$ and failed re-verification at doubled precision (residual 10^{-93} in place of 10^{-480}) — spurious.

The contrast with Theorem 1.1 is the point. In this basis the explicit formula splits as

$$\sqrt{h_m h_n}\,W_{mn} \;=\; \underbrace{2\sqrt{\pi}\,q_{mn}}_{\text{pole}} \;+\; \underbrace{\sqrt{2\pi}\,\big[\alpha_{mn} + \beta_{mn}(\gamma + \log 8\pi)\big]}_{\text{arch, rank 2}} \;-\; \underbrace{4\sqrt{\pi}\sum_{k\ge1} a_k\Big(-\tfrac{\zeta'}{\zeta}(1+2k)\Big)}_{\text{prime, rank}\,\ge 16},$$

with all of $q_{mn}, \alpha_{mn}, \beta_{mn}, a_k \in \mathbb{Q}$. The Γ-side of the basis carries a rank-two lattice; the primes are generic. The integrality of the approximation matrix is a purely archimedean phenomenon.

Remark 3.2 (The form is a Gram matrix of zeros)**.** Numerically $W_{00} = 2.7357 \times 10^{-8}$: not noise, but exactly $\sum_\gamma h_{00}(\gamma)$, dominated by the first zero ($e^{-\pi\gamma_1/2}$ with $\gamma_1 = 14.13\ldots$). The identity pole + arch − prime $= \sum_\gamma h_{mn}(\gamma)$ was confirmed to 39–56 digits on all sixteen entries. Thus W is the Gram matrix of the vectors $\big(\widehat{P}_m(\gamma_j/2)\big)_j$ under the weight $2(\gamma_j^2+\frac{1}{4})\,|\Gamma(\frac{1}{4}+\frac{i\gamma_j}{2})|^2$; its eigenvalue cascade (9.1, 5×10^{-4}, 6×10^{-10}, 3×10^{-18} on the even block) tracks the successive weights $e^{-\pi\gamma_k/2}$. On RH the positivity of W is trivial — $h_{mm} \ge 0$ on $\mathbb{R}$ — so the content of Weil positivity in this basis is exactly the reality of the γ_j, while the $O(1)$ pole/arch/prime structure cancels to eight digits before the first zero is heard.